

Lab 9: UART

Today we will work with the provided UART modules.

Discussion – How are we doing?

1. Any issues?

Task – Getting Familiar with the UART

1. Put the provided UART RX and TX into functional simulation, individually
2. Obtain full familiarity with the operation of the components.
 - a) Which ports are toggled when data is received on the RX?
 - b) Which ports must be toggled in order to transmit data on the TX?
3. Think about a state machine design to control the data flow using the UART components

Task – UART echo on the Pynq board

For this task, work in groups. Make sure that your group has at least one member with a UART/USB converter. This task is closely related to the functionality you will use in the next Assignment

1. Create a block diagram In Vivado that listens to data via the UART RX. Once the data is received, it should be sent via UART TX back to the computer similarly to an *echo* method.
 - a) Think about which components are needed. (*Hint: A finite state machine can help*)
2. Implement the system, verify in simulation.
3. Program the Pynq board, verify functionality.